Methods of Theoretical Physics

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Abstract

This course was taught by Dr. Claudio Chamon during the Spring 2025 semester at Boston University. We used the text Basic Training in Mathematics: A Fitness Program for Science Students by R. Shankar.

Contents

Chapter 1

Sets

Lecture 1: Syllabus Day

Tue 21 Jan 2025 12:30

The professor is recommending ChatGPT as a tutor. This is awesome. Problem sets due Thursdays at 11:59pm.

Prof asked "What are numbers?" and this chick answered "erm uh numebrs are uh do you want me to give the set theory definition? Erm its difficult to define numbers it requires complex set theory" lmfao. We will define numbers as an element of an algebraic structure, like a field or group. We take $0 \in \mathbb{N}$. We use the convention of writing $\mathbb{Z}_2$ instead of $\mathbb{Z}/2\mathbb{Z}$.

A cartesian product is a product in Set; an object $A \times B$ together with canonical projections that are final in the category $\mathsf{Set}^{A,B}$.

Lecture 2: lol i should drop this class

Thu 23 Jan 2025 12:31